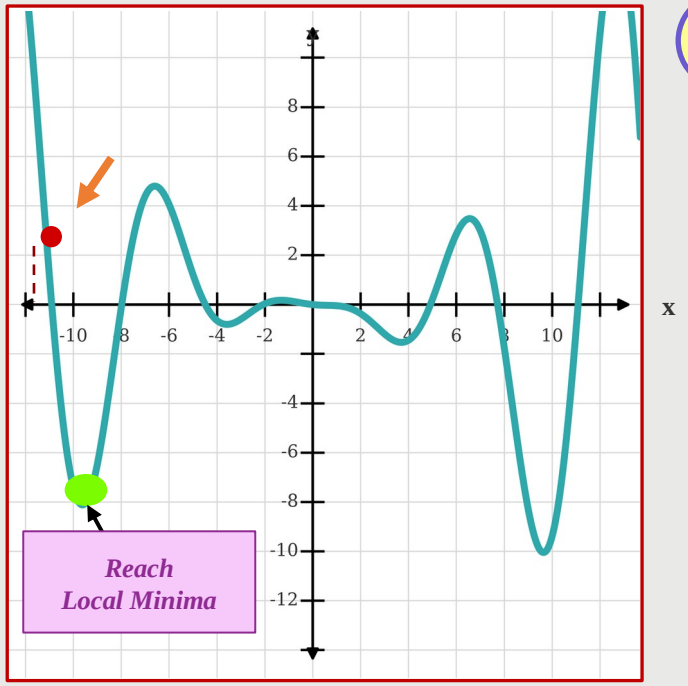
Non-Convex Function

May reach different minimum value based on:

initial point and learning rate

$$f(x) = \frac{x^2\cos(x) - x}{10}$$



step	0.1
initial X	-11
delta	0.1

X is in between -9.7 and color red bold -9.6

f(x) is in between -8.08 and -8.011

x	-11	-10.9	-10.8	-10.7	-10.6	-10.5	-10.4	-10.3	-10.2	-10.1	-10	-9.9	-9.8	-9.7	-9.6	-9.7	-9.6	-9.7
f(x)	1.15	-0.04	-1.19	-2.26	-3.27	-4.19	-5.03	-5.77	-6.41	-6.95	-7.39	-7.72	-7.96	-8.08	-8.11	-8.08	-8.11	-8.08
x-delta	-11.1	-11	-10.9	-10.8	-10.7	-10.6	-10.5	-10.4	-10.3	-10.2	-10.1	-10	-9.9	-9.8	-9.7	-9.8	-9.7	-9.8
f(x-delta)	2.39	1.15	-0.04	-1.19	-2.26	-3.27	-4.19	-5.03	-5.77	-6.41	-6.95	-7.39	-7.72	-7.96	-8.08	-7.96	-8.08	-7.96
x+delta	-10.9	-10.8	-10.7	-10.6	-10.5	-10.4	-10.3	-10.2	-10.1	-10	-9.9	-9.8	-9.7	-9.6	-9.5	-9.6	-9.5	-9.6
f(x+delta)	-0.04	-1.19	-2.26	-3.27	-4.19	-5.03	-5.77	-6.41	-6.95	-7.39	-7.72	-7.96	-8.08	-8.11	-8.05	-8.11	-8.05	-8.11
slope	-24.3	-23.4	-22.2	-20.8	-19.3	-17.6	-15.8	-13.8	-11.8	-9.8	-7.7	-5.7	-3.6	-1.5	0.3	-1.5	0.3	-1.5
next x	-10.9	-10.8	-10.7	-10.6	-10.5	-10.4	-10.3	-10.2	-10.1	-10	-9.9	-9.8	-9.7	-9.6	-9.7	-9.6	-9.7	-9.6